Instructions for the water rocket and safety measures

Before starting:

- Collect the water rocket and all the materials given to you
- Choose where to launch the water rocket. This is not to be launched indoors. Suggest areas are as follows –
  - i) An open field
  - ii) Lawn
  - iii) Garden

Assemble the rocket:

- Attach the nose cone to the plastic bottle. Make sure it connects tightly
- Secure the fins to the fin sleeve by glueing them together
- Attach the 3D printed fins to the end of the plastic bottle
- Additional glue can be used to ensure the rocket pieces are attached to the bottle

Launching the rocket:

- Check the bottle for any unwanted damage or leakage
- Fill the bottle with approximately 900ml of water
- Attach the pump nozzle to the cap of the bottle and secure it so there will be no leakage
- Place the rocket in the test area with the nose cone facing upwards
- Start pumping the water rocket. It's best to kneel down when pumping
- Keep pumping the rocket till it launches. Expect some water droplets on your body.
- Be careful not to lose any equipment during the process and while collecting the bottle.

Educational benefits:

Physics and Engineering concepts –

- Newton's Laws of Motion. For example, the first law of motion is used when the rocket remains at rest until launched. Concepts like F = ma as well is involved in the testing of the water rocket.
- Aerodynamics. The study of lift, drag and stability is essential during the water rocket.
- Thrust and pressure. These concepts determine the relationship between the compressed air and force which eventually launches the rocket.

Scientific Enquiry and Explanation –

- Hypothesis
- Data collection and analysis
- Use of scientific methods

Soft Skills Development –

- Teamwork
- Problem solving skills
- Project planning